

Question Paper

Exam Date & Time: 02-Jan-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.Tech. (COMPUTER SCIENCE & ENGINEERING) EXAMINATIONS - DECEMBER 2024 / JANUARY

2025

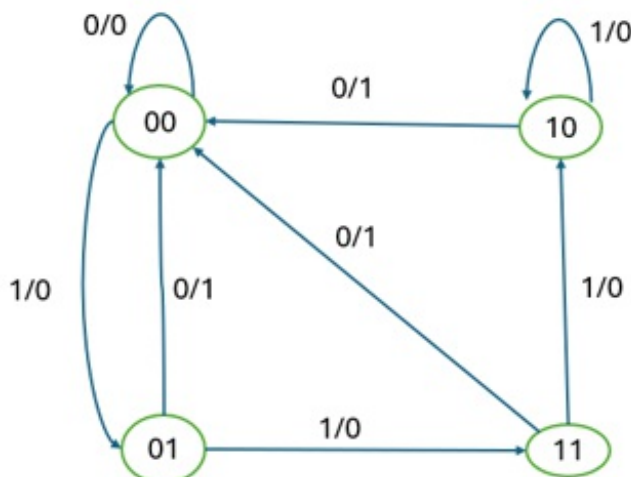
SUBJECT: CSE 2123/CSE_2123 - DIGITAL SYSTEM DESIGN
(MAKEUP)

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Design a synchronous sequential circuit for the following state diagram using JK Flip flop (5)



- 1B) Build the equations for low noise margin and high noise margin in logic circuits. Identify the terms present in the equations. Identify the types of power dissipation in logic circuits. (3)

- 1C) Simplify the following function using a 4-variable K-map, then implement the simplified expression using only NAND gates. (2)

$$F(A, B, C, D) = A'B'C'D + A'B'CD + AB'C'D + AB'CD' + ABC'D + ABCD'$$

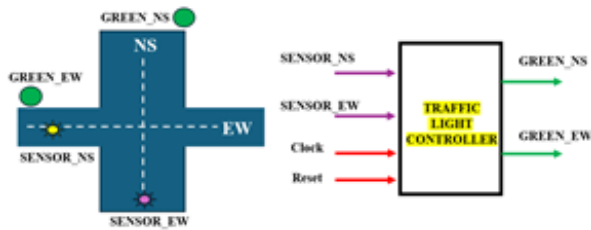
- 2A) Explain the structure and working of a Full Adder. How does it differ from a Half Adder in being more suitable for multi-bit binary addition? (5)

- 2B) Design and develop 4-bit carry-lookahead adder. Apply suitable approach to optimize the design to reduce the delay. (3)

- 2C) Develop a 4-to-2 priority encoder using Verilog HDL. Implement the circuit and simulate its output for different input combinations, ensuring that the highest priority input is encoded. (2)

- 3A) Design a traffic light controller for a four-way intersection. The intersection allows two directions of traffic: North-South (N-S) and East-West (E-W). To ensure safe passage, only one direction can have a green light at a time. The controller must follow these rules: (5)

Priority: The North-South direction has higher priority over the East-West direction.



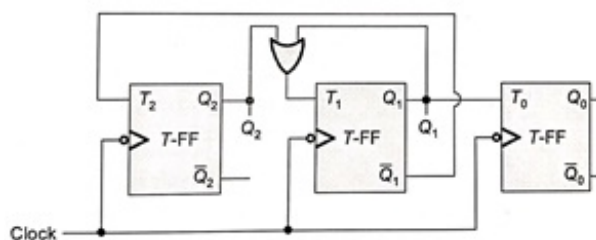
Traffic Requests: There are two sensors detecting vehicles waiting to cross in each direction: SENSOR_NS is HIGH if there is traffic in the North-South direction. SENSOR_EW is HIGH if there is traffic in the East-West direction. GREEN_NS and GREEN_EW are output signals that control the green light for each direction: GREEN_NS goes HIGH to turn the green light on for the North-South direction. GREEN_EW goes HIGH to turn the green light on for the East-West direction. If both SENSOR_NS and SENSOR_EW are HIGH, GREEN_NS (North-South) should be HIGH due to the priority rule, and GREEN_EW should stay LOW. When the green light is ON for a direction, it should remain green until the traffic in that direction clears (i.e., the respective request signal goes LOW).

Design an Finite State Machine for the Traffic Controller and synthesize it. Assume missing data/ conditions if any and state them.

- 3B) State the comparison between combinational and sequential circuits. Design a new flipflop whose characteristics are given below, using T- flipflop. (3)

A	B	Q_{n+1}
0	0	0
0	1	1
1	0	$\overline{Q_n}$
1	1	Q_n

- 3C) Draw an Algorithmic State Machine (ASM) chart to count the number of Ones present in a number input to the system. (2)
- 4A) Design a simple processor with a control circuitry that can perform the addition/subtraction of two 4-bit data. Provide suitable explanation. (5)
- 4B) Design a 3-bit synchronous counter using D-flipflops that counts even states only and repeats the sequence after reaching the maximum even number of the state. Consider the counter starts from the '0' state. Assume unused states are don't cares. (3)
- 4C) Write the Verilog behavioral description of a four-bit up-down counter with parallel load using three control inputs for the three operations Load, Up, and Down. The order of precedence is Load, Up, and Down. (2)
- 5A) Since a computer has many registers, paths must be provided to transfer information from one register to another. An efficient scheme for transferring the information between registers in a multiple-register configuration is a common bus system. Design a 4-bit common bus system for 4-registers of 4-bits each using multiplexers. (5)
- 5B) Design a synchronous 3-bit up-down counter using JK-flipflops. Draw an appropriate sequence diagram, excitation table, and logic diagram to realize the said counter. (3)
- 5C) Determine the number of unused states and MOD of the following counter: (2)



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